

LEDGER · 2026-09-04 · AGTECH / MATERIALS SCIENCE

Paraffin-Functionalized Sand (PFS) Mulch via Hexane-Solvent Evaporation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Low-Density Polyethylene (LDPE) Black Mulch Film
Entry application	high-value horticultural crops in arid regions
Measured advantage	none
Time to first revenue	18 months
Gross margin at parity	95.1%
Startup CapEx	\$465,000
Payback from first sale	0.6 months
Capital productivity	28.65x
Regulatory risk	Moderate — qualification costs reported (\$150,000 estimated)

Why it failed

- Voided by: Mismatched comparison, Economics undisclosed, Negative control / sub-2x, Form-factor / handling regression, Absence search missing
- Absence asserted, search unreported

Assessment

The report explicitly concedes complete commercial failure. A severe material mass penalty of 60 metric tons of sand per hectare versus 100 kg of LDPE film creates insurmountable cost, labor, and logistical barriers. Startup CapEx (\$465,000) exceeds the threshold and no structured market absence ledger is provided.

Also flagged

- runsheet_no_quantities

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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